
CSV To ScriptableObject



This powerful tool lets you seamlessly convert **CSV files into ScriptableObjects**, making data management and manipulation in your Unity projects effortless. 📄 ➡ 📦

Use it to easily **create any new ScriptableObjects** or **update any existing ones** with just a few clicks. 🛠️

Streamline your workflow and supercharge your productivity. 🔥

April 30, 2025

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1 Why Use This Tool?

Transform your Unity workflow with this intuitive tool designed to revolutionize how you handle ScriptableObjects through CSV files.

Say goodbye to tedious manual data entry - whether you're managing thousands of game items, character stats, or level configurations, this tool handles it all with just a few clicks.

2 Key Capabilities

 **Editor Mode Compatible** This utility operates exclusively within Unity's Editor environment and cannot modify or create ScriptableObjects during runtime.

Core Functionality:

- Create and update ScriptableObjects using precise field mapping from local CSV data sources
- Seamless compatibility with any custom ScriptableObject implementation
- Preview system for data mapping of CSV columns

Benefits

- You are not bound to any specific CSV format. 
(You can customize the CSV format to your needs.)
- You can use any CSV file to generate large amounts of data in your project. 
- You can update existing ScriptableObjects with new data from CSV files. 
(Created ScriptableObjects does not need to be created from this tool.)
- Configure precise update operations with flexible ID matching that supports multiple fields and conditional logic between ScriptableObject and CSV data sources 
- Supports inherited ScriptableObjects. 

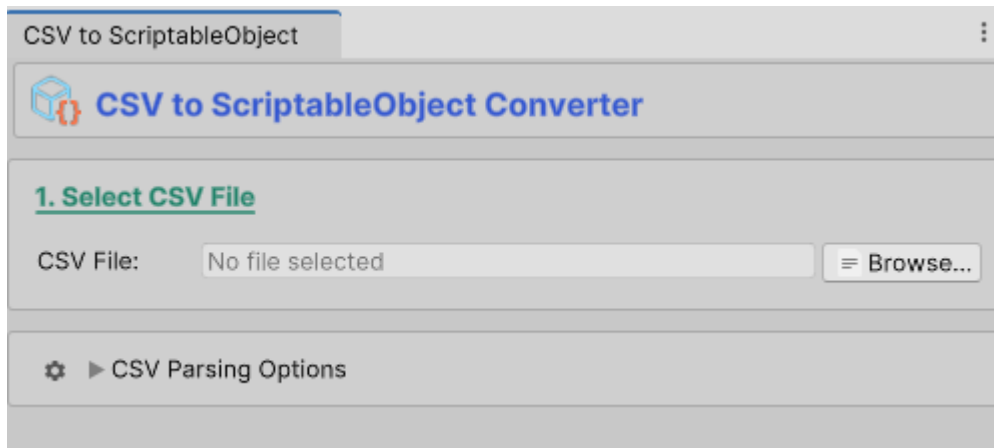
3 🚀 Setup

3.1 🛠️ Open the Tool

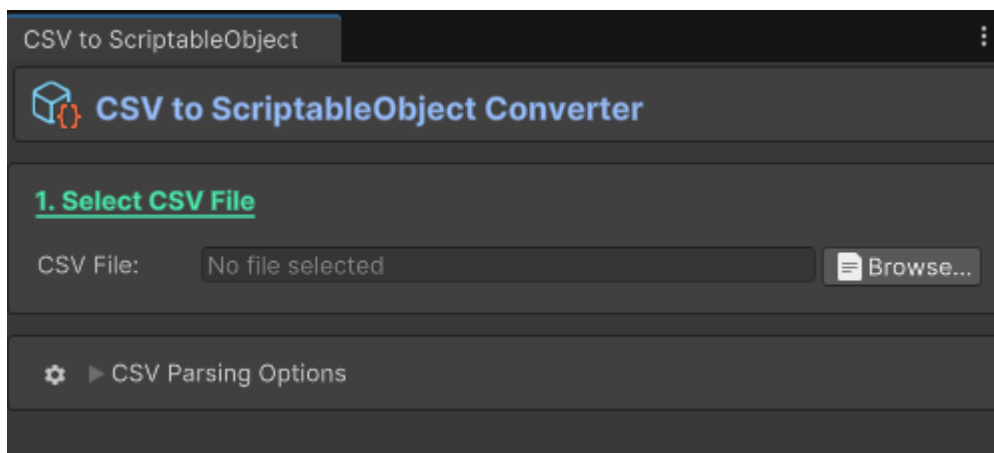
To get started, go to the Unity menu: **“Tools”** → **“CSV To ScriptableObject”**.

This will open the tool window.

Light Mode Preview:



Dark Mode Preview:

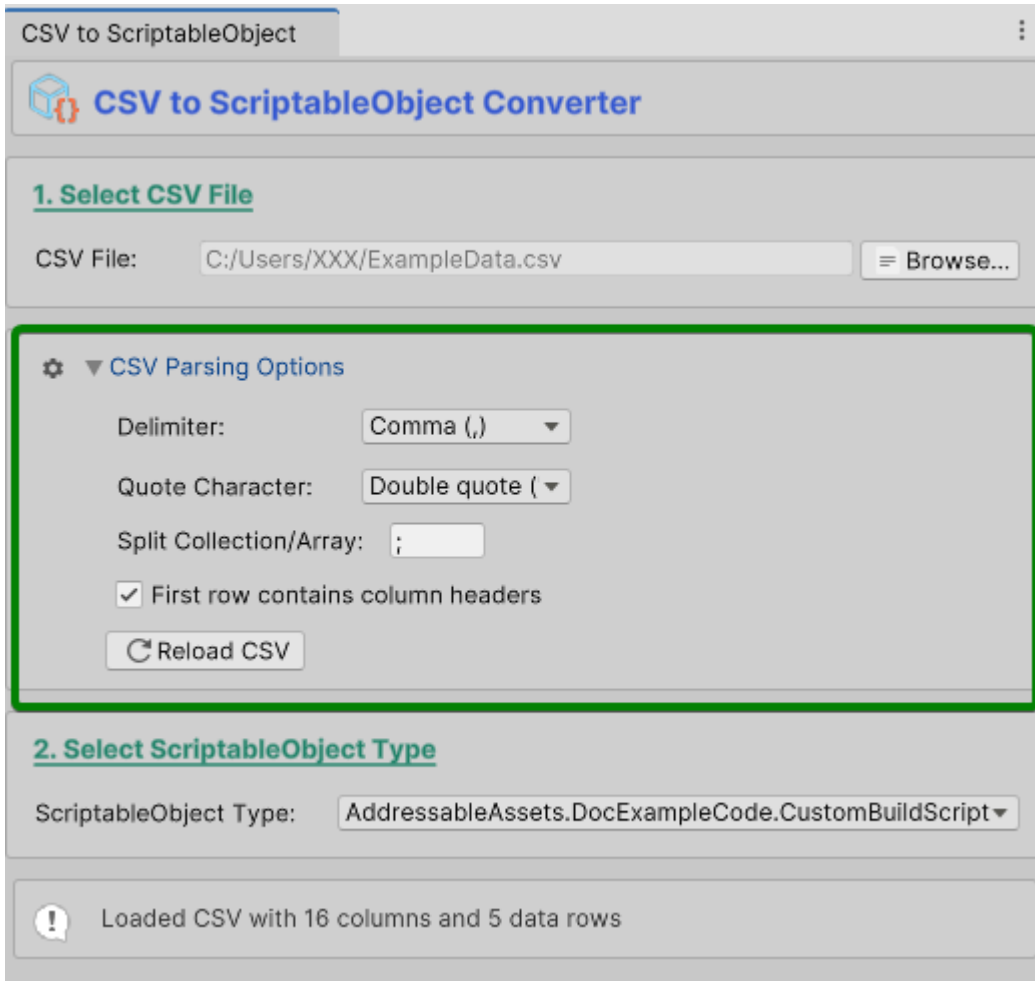


i Info: Light mode is used in this documentation for better readability.

3.2 Select the CSV File

Expand the “**CSV Parsing Options**” to access all CSV-related settings.

These settings need to be configured before you load the CSV file.



CSV to ScriptableObject

CSV to ScriptableObject Converter

1. Select CSV File

CSV File:

⚙️ ▼ CSV Parsing Options

Delimiter:

Quote Character:

Split Collection/Array:

☒ First row contains column headers

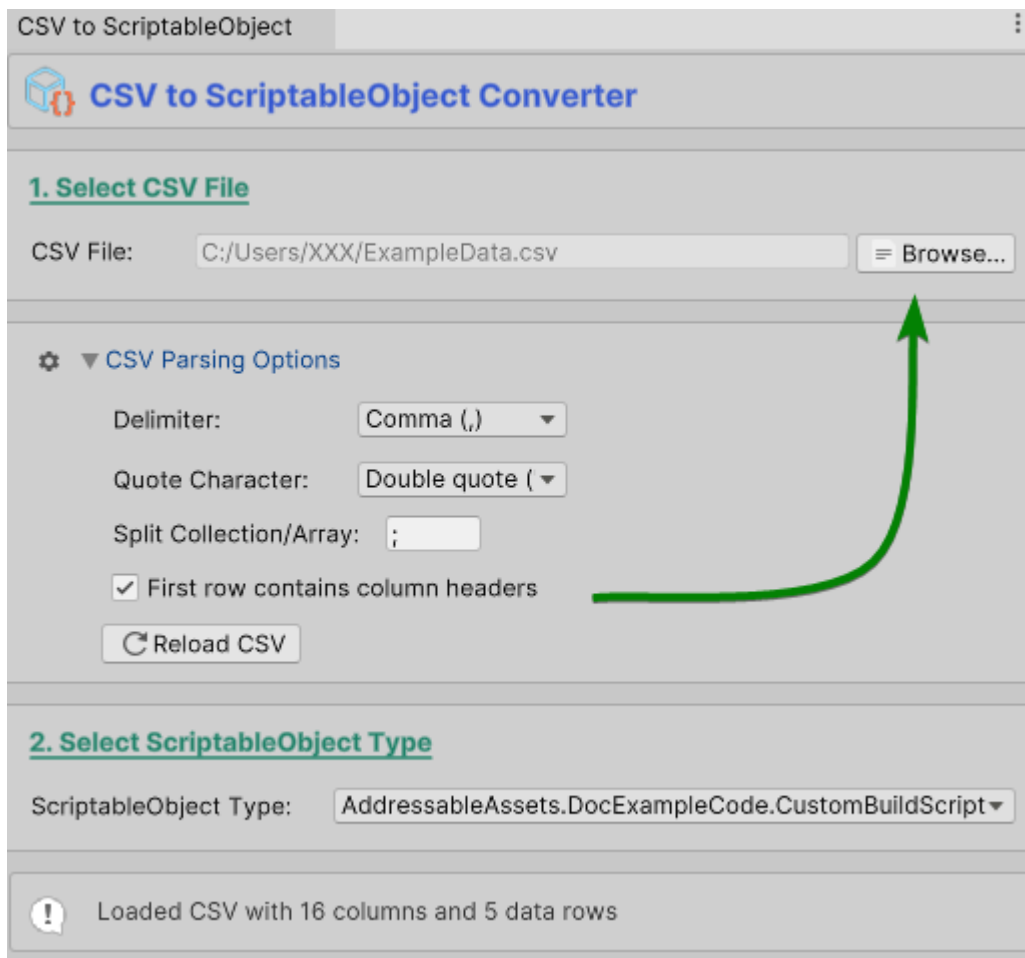
2. Select ScriptableObject Type

ScriptableObject Type:

! Loaded CSV with 16 columns and 5 data rows

After you configure the settings, you can load the CSV file.

Click on button “**Browse...**” to select the file you want to import. 📁



CSV to ScriptableObject

CSV to ScriptableObject Converter

1. Select CSV File

CSV File:

CSV Parsing Options

Delimiter:

Quote Character:

Split Collection/Array:

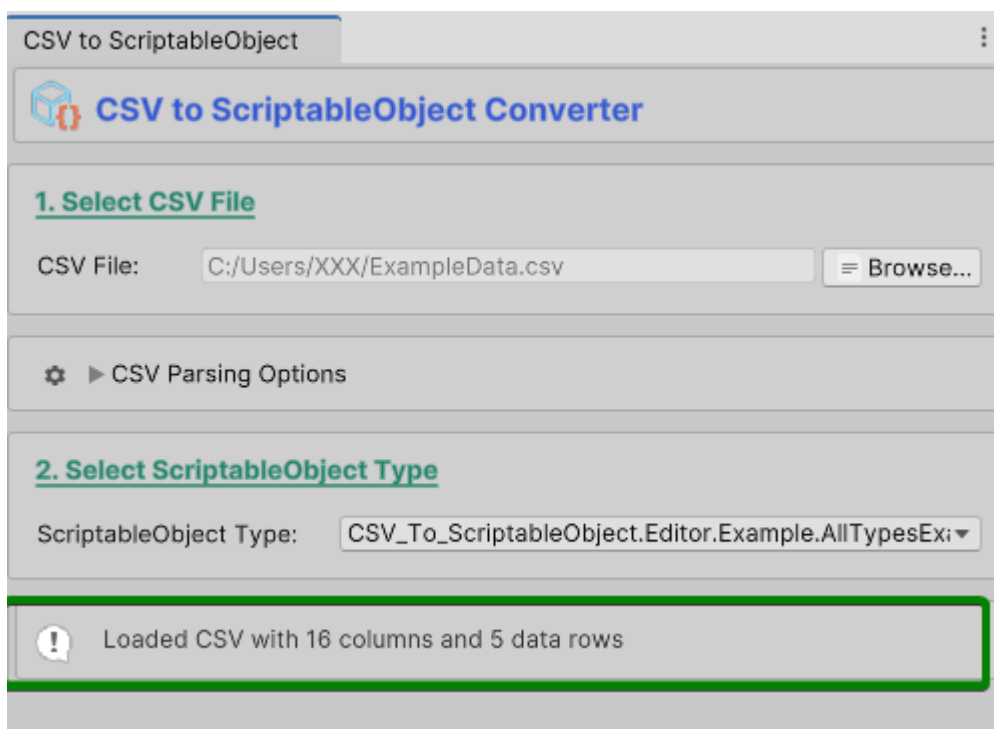
☒ First row contains column headers

2. Select ScriptableObject Type

ScriptableObject Type:

! Loaded CSV with 16 columns and 5 data rows

Once selected, the tool will **automatically detect the number of columns and rows** in your file. 🔍



CSV to ScriptableObject

CSV to ScriptableObject Converter

1. Select CSV File

CSV File:

CSV Parsing Options

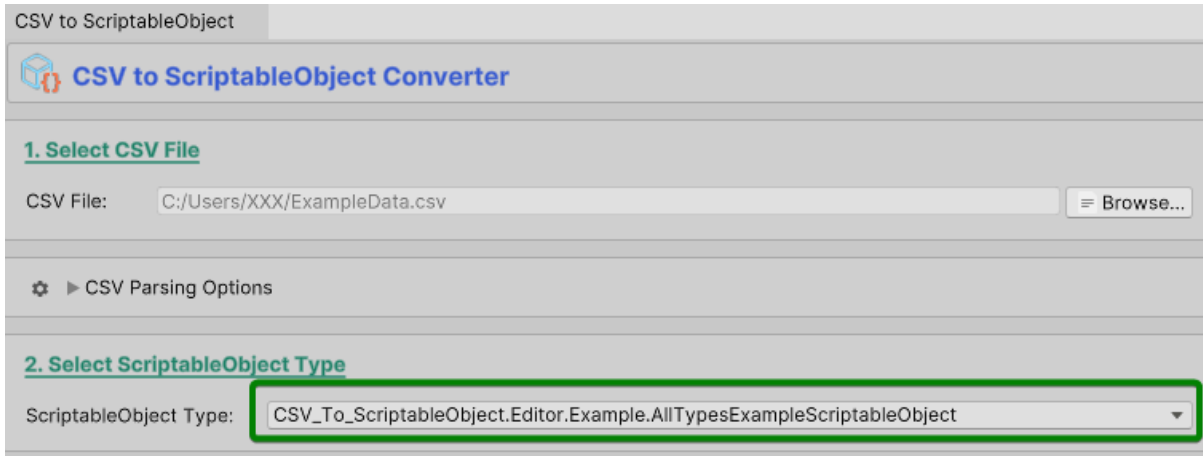
2. Select ScriptableObject Type

ScriptableObject Type:

! Loaded CSV with 16 columns and 5 data rows

3.3 Select the ScriptableObject to Create or Update

From the dropdown, **select the ScriptableObject** you want to **create or update**.



CSV to ScriptableObject

CSV to ScriptableObject Converter

1. Select CSV File

CSV File:

⚙️ ▶ CSV Parsing Options

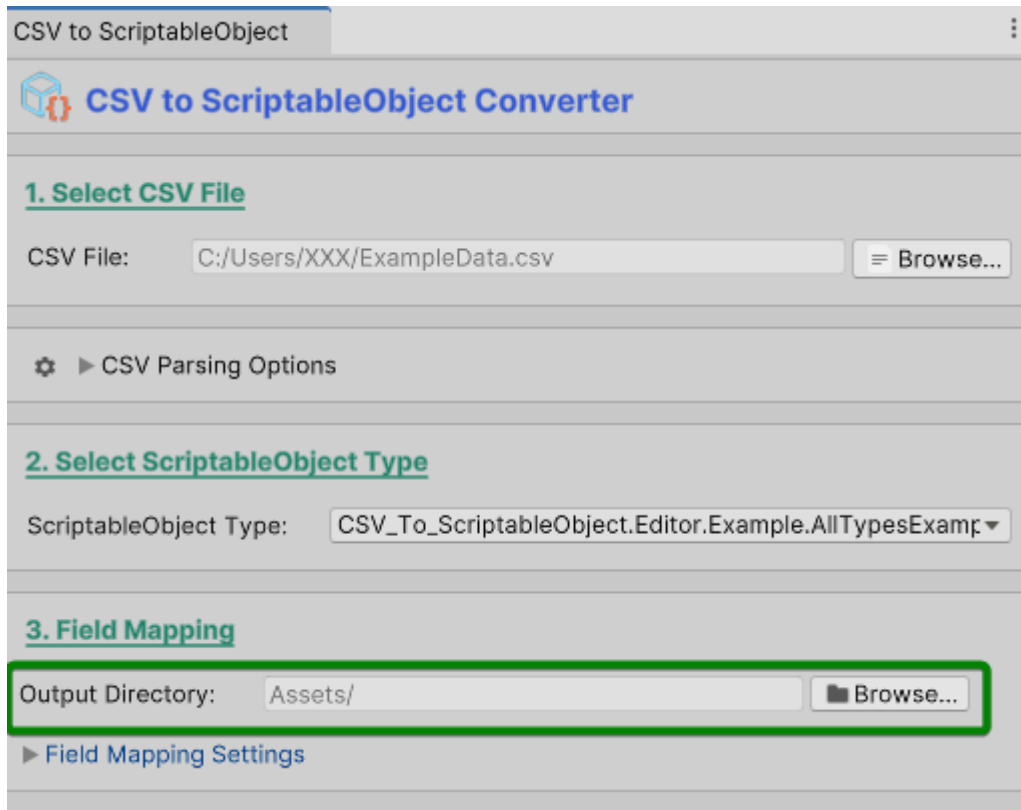
2. Select ScriptableObject Type

ScriptableObject Type:

 Warning: The **ScriptableObject** must include a **public constructor** to enable creation of new ScriptableObjects.

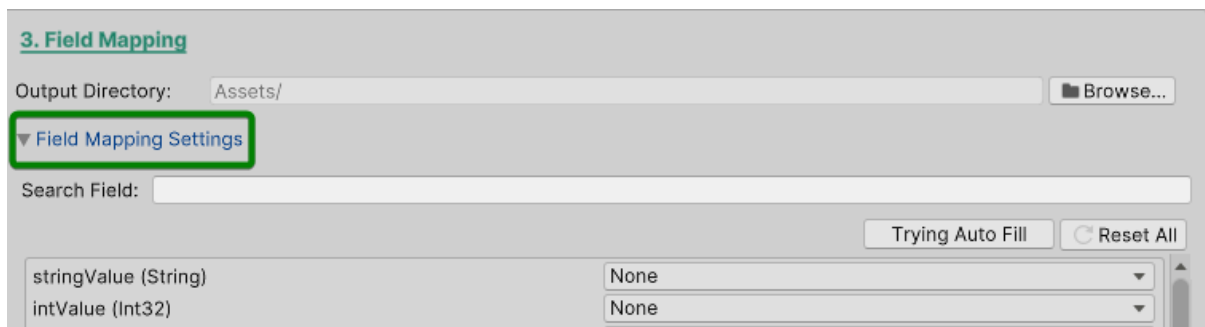
3.4 Map CSV Columns to ScriptableObject Fields

First, **select the output directory**  where the ScriptableObjects will be **created**  or **updated** .



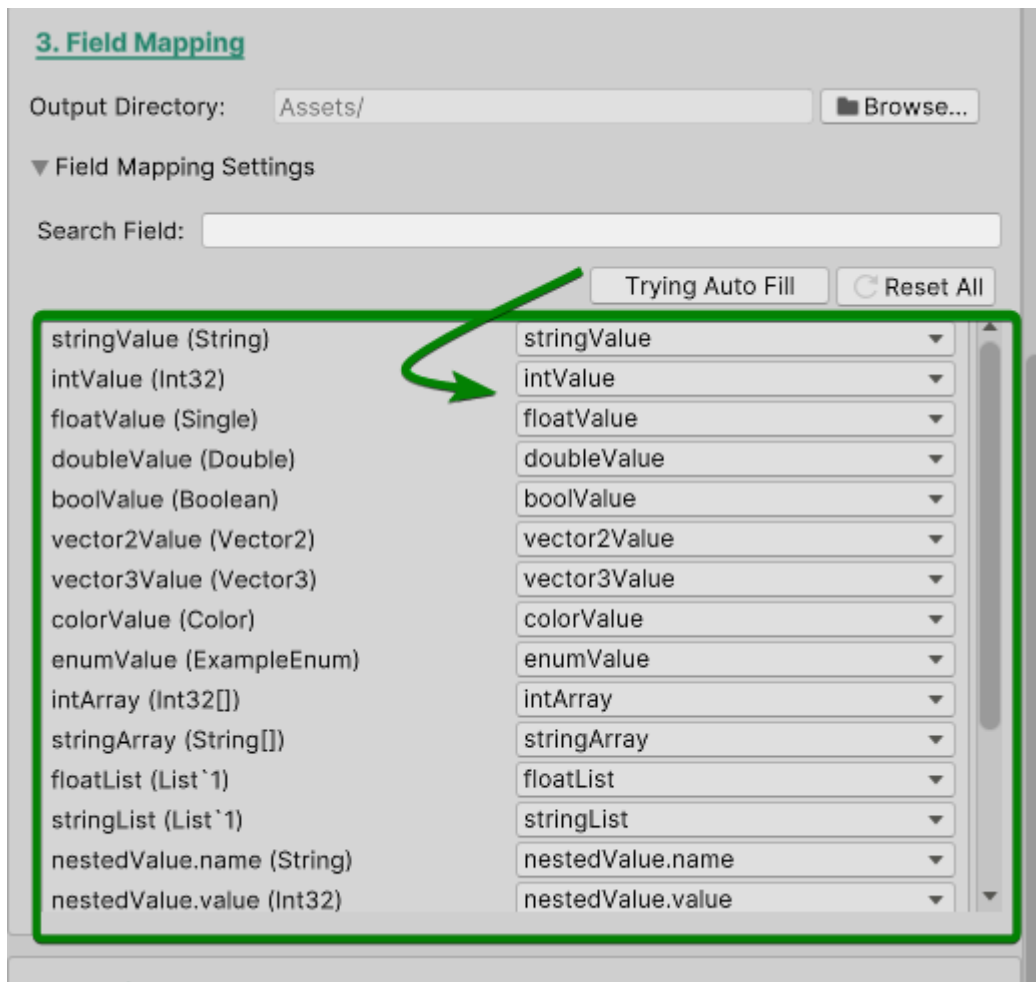
The screenshot shows the 'CSV to ScriptableObject Converter' window. It has three main sections: '1. Select CSV File', '2. Select ScriptableObject Type', and '3. Field Mapping'. In the '3. Field Mapping' section, the 'Output Directory' field is highlighted with a green box. The field contains the text 'Assets/' and has a 'Browse...' button next to it. Below the 'Output Directory' field is a link for 'Field Mapping Settings'.

Then expand “Field Mapping Settings” and **map the CSV columns to the ScriptableObject fields** by linking each CSV column to the corresponding field. 



The screenshot shows the 'Field Mapping Settings' section. It has a search bar labeled 'Search Field:'. Below the search bar, there are two columns of fields. The first column contains 'stringValue (String)' and 'intValue (Int32)'. The second column contains 'None' and 'None'. There are buttons for 'Trying Auto Fill' and 'Reset All'.

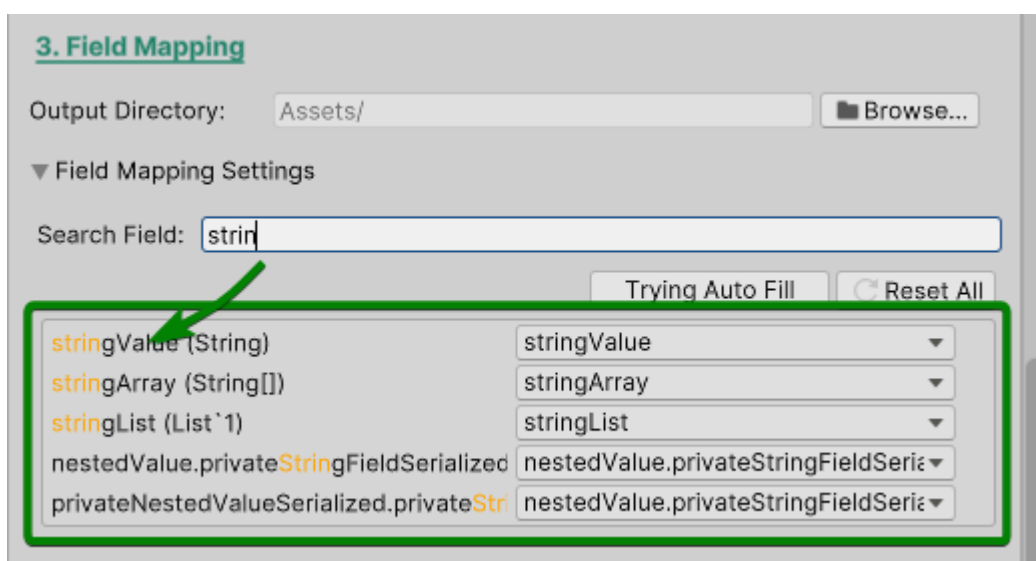
i Tip: You can use the **“Try Auto Fill”** button to automatically match columns to fields. 🤖
(Recommended to use before manual mapping.)



You can manually adjust any mapping as needed. ✎

It's entirely up to you which fields to process and which to ignore.

i Tip: Use the **search bar** to quickly find specific fields when dealing with many fields. 🔍

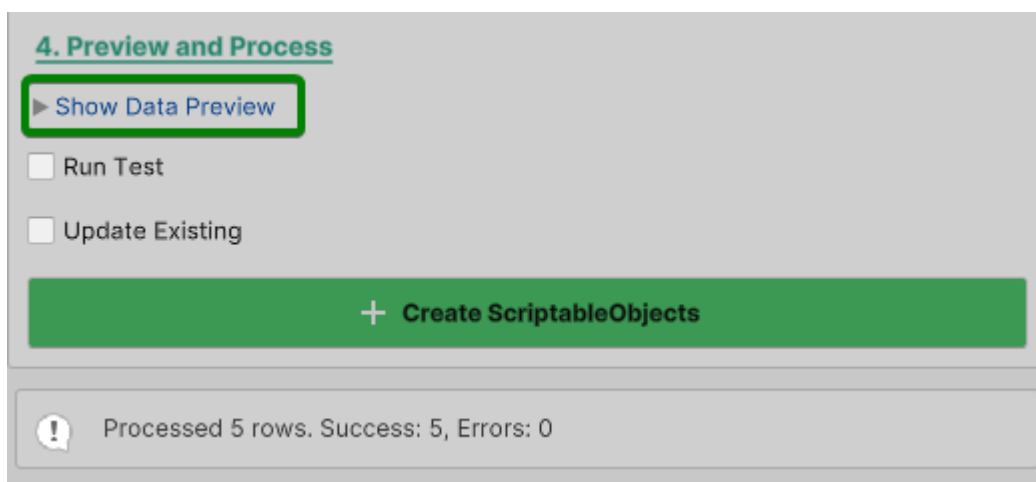


3.5 Create or Update ScriptableObjects

i Tip: Collapse previously configured sections “Field Mapping Settings” for a clearer view of the current step.



Before proceeding, you can check the “**Preview**” to review how the data is mapped. [↗](#)



After expanding the **“Preview”** section, you can see how the mapping looks like for CSV file.

Preview show a few rows of the CSV file and informate you if you missed any field mapping. ⚠

You are not required to map all fields, but it's only to inform you if you missed any field mapping.

4. Preview and Process

▼ Show Data Preview

<< < Page 1/31 > >> Rows per page: 15 Showing rows 1-15 of 458

⚠ player_id (not mapped)	✓ username → privateNestedValueSerialized.name → nestedValue.name	⚠ level (not mapped)	⚠ game_ (not map
1	hbandt0	46	3518
2	kmilnthorpe1	53	7667
3	aha2	19	4070
4	fmcwhirter3	39	8545
5	rleopard4	86	3184
6	mellerey5	68	6311
7	alarciere6	47	557

Enable the **“Run Test”** checkbox to verify your mapping by processing a small sample of the CSV data first, before large processing. 🧪

4. Preview and Process

► Show Data Preview

☒ Run Test

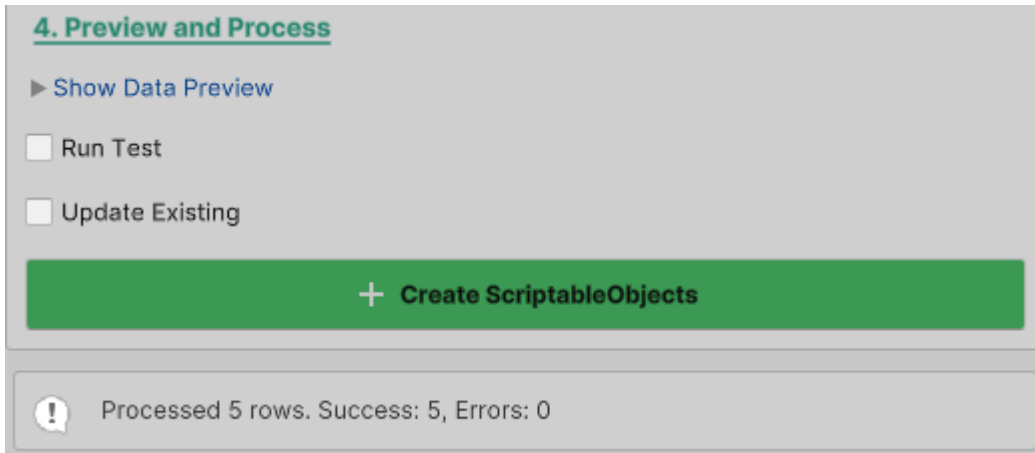
☐ Update Existing

+ Create ScriptableObjects

This helps ensure everything is set up correctly before running the full process. ✓

3.5.1 Create New ScriptableObjects

If you're ready just click the **“Create”** to **generate new ScriptableObjects**.



4. Preview and Process

► Show Data Preview

☐ Run Test

☐ Update Existing

+ Create ScriptableObjects

! Processed 5 rows. Success: 5, Errors: 0

Once the process finishes, a **confirmation message** will appear. 

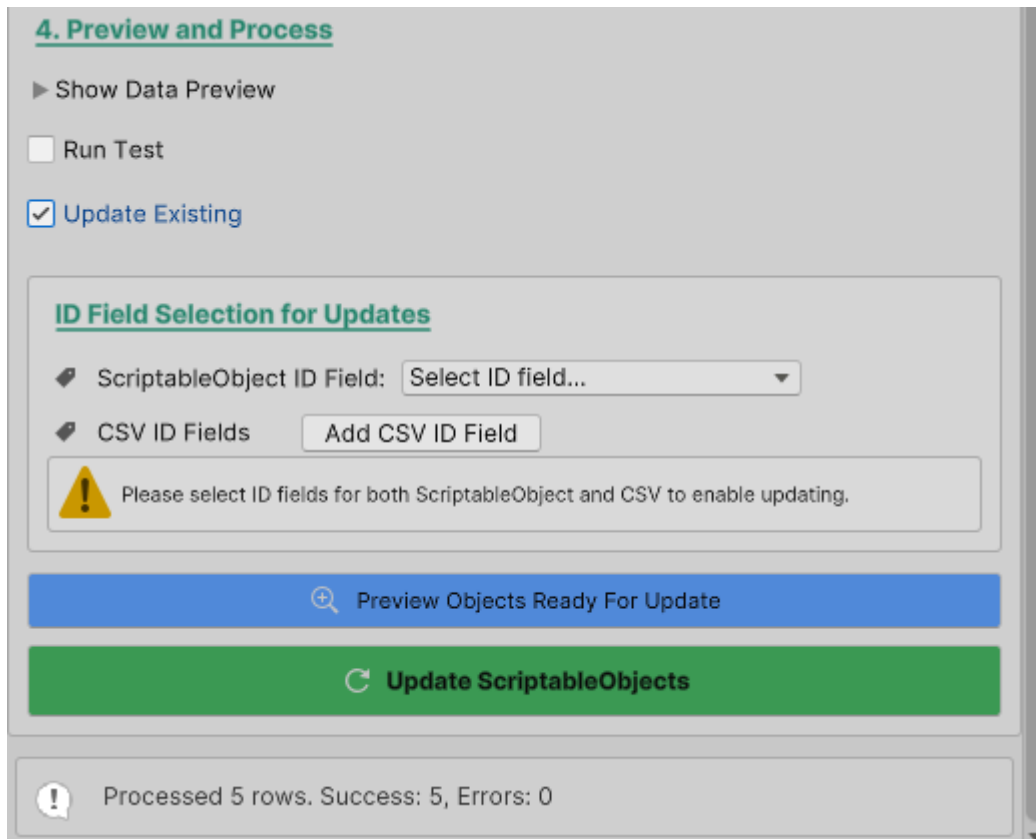
A detailed log is also available in the Unity console for verification. 

Created ScriptableObjects will be named based on the **“File Name”** field in the CSV file with row number

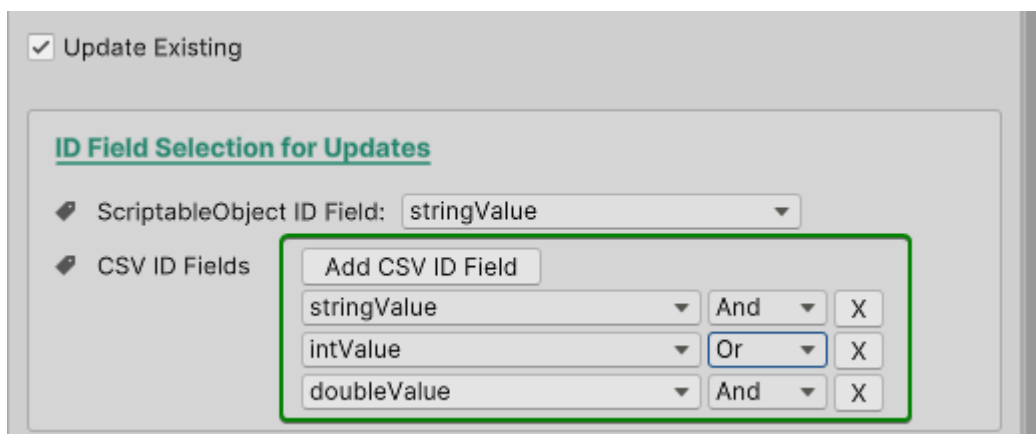
3.5.2 Update Existing ScriptableObjects

⚠ Important: Back up your data to avoid any potential data loss before proceeding.

To **update existing ScriptableObjects**, enable the “**Update Existing**” checkbox.



When updating, **specify matching ID field/s** from the ScriptableObject and CSV columns for proper matching.  

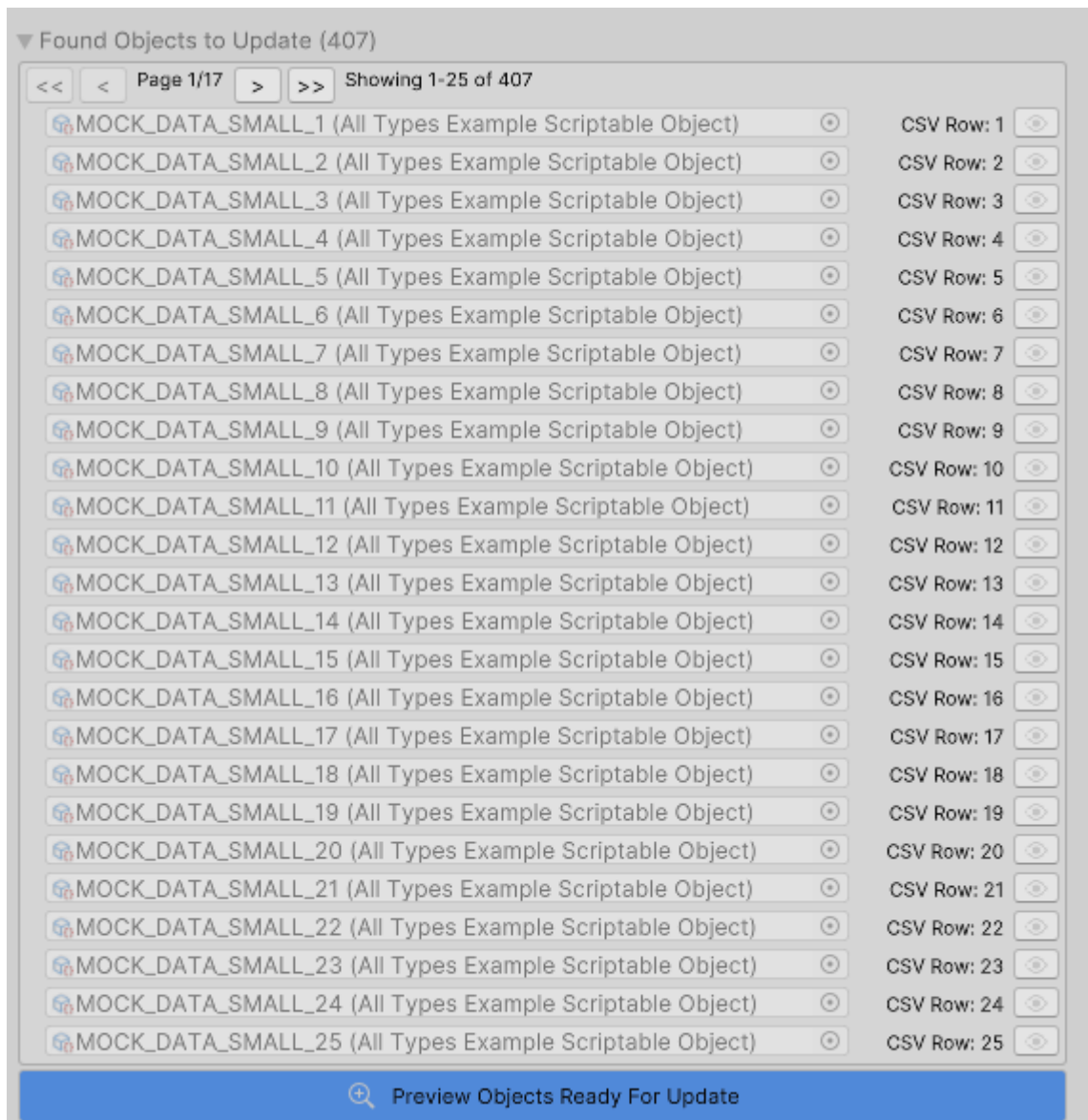


CSV fields allow you to select multiple ID fields.

You can define whether the matching should use **AND** or **OR** logic, depending on your needs.

The last condition is reserved for the next comparison and will not be applied.

Click **“Preview”** to see which **ScriptableObjects** match your criteria and will be updated.

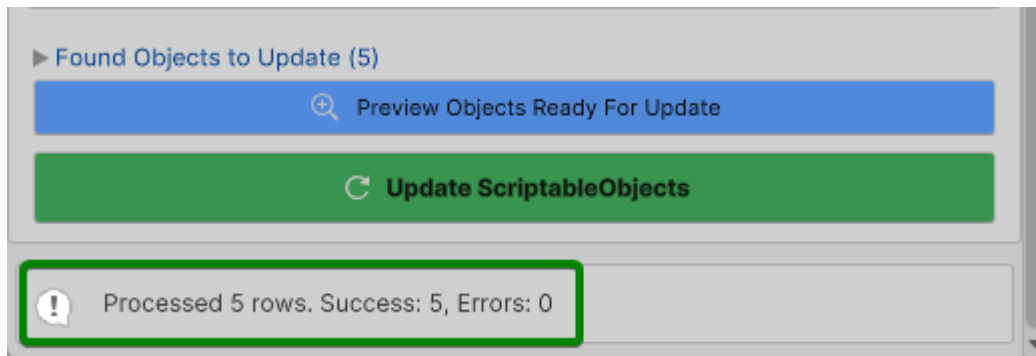


Preview displays a read-only view of the ScriptableObjects that will be updated

Helps you detect any duplicate entries with the same ID field.

If you're ready just click the **“Update”** to start **updating existing ScriptableObjects**.

The tool skips CSV rows that don't match any existing ScriptableObjects, only updating found matches.



4 ? FAQ

4.1 Why CSV?

CSV (Comma-Separated Values) is a widely used format for data storage and exchange.

It is simple, human-readable, and can be easily edited in spreadsheet applications like Microsoft Excel or Google Sheets.

This makes it an ideal choice for managing large datasets, especially when working with Unity.

CSV format enables efficient management of extensive datasets, and this tool seamlessly integrates such data into your Unity project workflow.

4.2 Why convert CSV to ScriptableObjects?

ScriptableObjects are a powerful feature in Unity that allows you to create reusable data containers.

They are ideal for storing large amounts of data, such as game settings, character stats, or level configurations.

4.3 Can I use this tool to update existing ScriptableObjects?

Yes, this tool allows you to update existing ScriptableObjects with new data from CSV files.

You can specify matching ID fields to ensure that the correct ScriptableObjects are updated with the new data.

You can also define specific fields to update, so you can keep the rest of the data intact.

These ScriptableObjects do not need to be created from this tool.

4.4 Can I use this tool to create new ScriptableObjects?

Yes, this tool allows you to create new ScriptableObjects from CSV files.

4.5 Can I use this tool to create ScriptableObjects from multiple CSV files?

While the tool processes one CSV file at a time, you can use it repeatedly with multiple CSV files to achieve your data integration needs.

Simply repeat the process for each CSV file you want to

4.6 Does it support inherited ScriptableObjects?

Yes, this tool supports inherited ScriptableObjects, allowing you to create or update derived scriptable objects as well.

4.7 Can I use this tool to create or update ScriptableObjects at runtime?

No, this tool is designed to work exclusively within the Unity Editor and cannot modify or create ScriptableObjects during runtime.

This ensures that your data management process is efficient and reliable during development.

Creating or editing ScriptableObjects at runtime would require serializing/deserializing assets and writing to disk during gameplay, which could cause performance issues, file system security concerns, and potential data corruption.

Additionally, ScriptableObjects are designed as development-time assets in Unity's architecture. For runtime data modification, consider using other data structures like regular C# classes or PlayerPrefs.

4.8 🛠 Troubleshooting Guide

📁 Output Directory

- ✓ Directory exists in Assets folder
- ✓ Write permissions granted
- ✓ Valid path characters used

4.8.1 📦 Creation Issues

When creating ScriptableObjects fails, verify:

🔧 Class Configuration

- ✓ Public constructor exists
- ✓ Inherits from ScriptableObject

4.8.2 🔄 Update Issues

When updating ScriptableObjects fails, check:

📋 Data Validation

- ✓ ID fields properly configured
- ✓ Target objects exist

💡 If issues persist after verifying all points, check Unity console for detailed error messages.